



Review

Capacity management for hospitality and tourism: A review of current approaches

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ABSTRACT

In this article we provide a review of current capacity management approaches applicable to hospitality and tourism enterprises. While many of the traditional methodologies involve closed-form analytical calculations, newer methods have been developed that address more complex problems. These more recent methods include the creative use of mixed models such as those that integrate visitor-preference and operations-research models, linear programming, and simulation models. We consider the full range of methodologies, the advantages and disadvantages, and potential applications to both physical and human capacity estimation. We highlight the ways that various leisure industries can benefit from different approaches to capacity management.

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1. Introduction

Many leisure enterprises require substantial capital investment and expenditure. A single amusement park ride can cost more than the production cost of the blockbuster movie. For example, the Amazing Adventures of Spider-Man ride, which opened at Universal Studios' Islands of Adventure in Florida, cost US\$100 million (Sens, 2003). Similarly, a cruise ship in the Caribbean

Princess class recently cost approximately US\$500 million, with a berth cost of US\$200,000 (Jarossy, 2006). The value of a 150-room hotel can be as much as £13.6 million, with a cost per square meter of £1300 (Ransley and Ingram, 2001). Yet enterprises often make important capital investment decisions without taking advantage of the full range of sophisticated quantitative tools developed in the field of operations management. The capacity models developed by these researchers thus represent a valuable and often untapped resource for firms in the tourism and hospitality industries.

Capacity management has been referred to as “demand management” (Crandall and Markland, 1996) or “managing capacity and demand” (Fitzsimmons and Fitzsimmons, 2004). However, capacity and demand management are, in fact, distinct

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concepts. Demand management attempts to influence when and how many visitors attend or use a service (Taylor, 1980); capacity management ensures that sufficient capacity exists to meet this demand (Klassen and Rohleder, 2002). While capacity management has generally fallen within the domain of operations management, and demand management within the domain of marketing, intersecting methodologies such as yield and revenue management, the partitioning of visitors (by length of stay), and/or the adjustment of visitor participation levels rely on expertise from both of these functional areas to work effectively (Klassen and Rohleder, 2002). In this review, we are more concerned with how to adjust capacity to meet demand while also considering those aforementioned methods where demand adjustments cannot be disentangled from capacity management.

Capacity decisions, one of the most significant strategic operational concerns, affect how well an enterprise responds to the market today and in the future (Getz, 1987; Johnson, 1993; O'Reilly, 1986). Previous research on capacity management spans a continuum from limited studies on industry-specific capacity planning to multiple articles focusing on a specific methodology such as yield management, which has been broadly applied to diverse hospitality industries (Jones and Lockwood, 1998). However, the research literature suffers from a dearth of analysis exploring the full range of capacity management approaches applicable to multiple enterprises.

The development of capacity strategy is a key operational function for all leisure-related enterprises (Bozarth and Handfield, 2006; Davis and Heineke, 2005; Fitzsimmons and Fitzsimmons, 2004). The extent to which capacity satisfies demand has an impact on visitor experience, employee satisfaction, profitability, and long-term sustainability of both the resources and the enterprise itself. In many leisure sectors, excess capacity not only underutilizes the workforce and other physical resources, but also leads to increased waste and price reductions to stimulate demand (Davis and Heineke, 2005, p. 387). Inadequate capacity, on the other hand, can impair the visitor experience through degradation of facilities, overuse of natural resources, crowding, and increased waiting time. It can allow opportunities for competitors to enter the market and can overtax the workforce, leading to employee burnout and increased turnover. Fortunately, firms can borrow a number of ideas from other industries to facilitate better matching of supply to demand in their particular enterprise.

The purpose of this paper is to provide an overview of the available capacity approaches at an enterprise rather than a destination level. We begin by defining the capacity methodologies and focus for different enterprises. We then review the different capacity indicators and performance measurements that have been proposed in the literature. We also review the strategic (long-term) physical and labour capacity decisions and quantitative approaches, as well as the approaches that support physical and labour capacity in short-term planning horizons. Finally, we discuss the implications of these approaches for management, and we outline future research opportunities.

2. Capacity management decisions

For this paper, we separate capacity into two distinct perspectives, the strategic or long-term perspective and the tactical or short-term perspective. Strategic capacity decisions are made during the planning stages for a project as managers consider macro-level responses to existing and potential future demand (Porter, 1996; Schroeder, 2006; Davis and Heineke, 2005). Such decisions may include the number of hotel rooms an area can or should support; the land space, energy, and water required for a project; the available labour and skills; the anticipated overall size of the enterprise in terms of parking, seating, and production

requirements (in food services); and/or the carrying capacity issues for natural resources. Later in the planning process, the capacity focus shifts to the set of short-term actions taken to fulfill the planned strategies, often called the “tactics” (Schroeder, 2006). Examples of these decisions include the determination of the number of employees needed to meet peak demand during a summer lunch period and the best mix of table configurations to accommodate dinner demand given different party sizes. Long versus short term decisions are distinctly different, and the methodologies for addressing these problems often involve alternate approaches (see Fig. 1).

2.1. Long versus short term decision focus

In the strategic capacity planning phase, enterprises make decisions to ensure achievement of their desired objectives based on an assessment of their current situation or position, capabilities, shortcomings, and overall competitive position; the alternatives and risks involved; and the timing (Porter, 1996). Strategic capacity decisions include capacity size and expansion, carrying capacity or optimal use of the resource, and capacity flexibility. Capacity size refers to the maximum physical size of a facility or the optimal size of a workforce. Carrying capacity is the desired or optimal level of utilization of a given resource; beyond that capacity, negative impacts exceed levels specified by evaluative standards such as those for physical deterioration or minimum quality of recreational experience (Lawson and Baud-Bovy, 1977; Mathieson and Wall, 1982). Capacity flexibility represents the ability to respond to fluctuating demand using additional labour or adjustments to physical space. These strategies address the amount and timing of capacity changes and types of facilities needed for the long run (Schroeder, 2006). For physical facilities, capacity strategy is part of the total operations strategy; it is not merely a series of incremental capital-budgeting decisions. Both physical and labour capacity strategies include proactive planning for future growth, reactive response to existing demand, or a mix of these two strategies (Davis and Heineke, 2005, p. 388).

Short-term capacity management comprises the set of actions taken to fulfill the firm's strategy—specifically, the methods of implementation or the requirements for the strategic plan to take effect. Here, the focus shifts to relatively small time durations and specific processes (for instance, diners or visitors served per hour, rooms cleaned per shift, or skiers per day). Managers and planners are concerned with meeting demand in real time by means of such

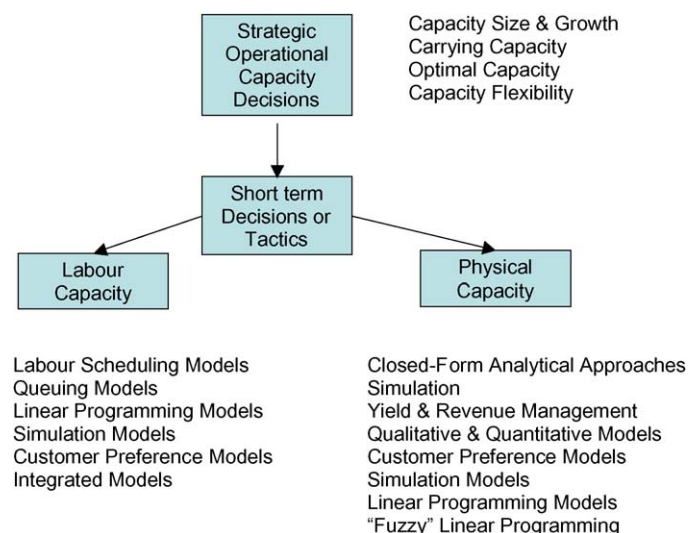


Fig. 1. Capacity decisions.

things as labour scheduling, flexible seating or partitioning, or opening up more seating capacity at lower price points as an event or departure date approaches. Other decisions address the design of the service process, including the service interaction time (the average amount of time an employee spends with a visitor); the level of visitor participation or self-service; the capacity cushion (extra space or labour added as a contingency); and queue reservations or partitioning, such as Disney's Fast-Pass program or Universal's VIP pass for special fast lines. The appropriate focus for these capacity decisions depends on the overall size and type of leisure enterprise.

2.2. Capacity resource focus

It is useful to view leisure-related enterprises in terms of the service process matrix, which was originally developed by Schmenner (1986) for the field of operations management. Using this framework, we classify enterprises according to their degree of labour intensity versus degree of interaction/customization (Table 1). Labour intensity is defined as the ratio of labour cost incurred to the value of the facilities and equipment. Generally, highly labour-intensive businesses require relatively little capital expenditure for facilities and equipment, while businesses with low labour intensity have low levels of labour cost relative to costs for facilities and equipment. The degree of interaction and customization reflects the level of service customization for the visitor and the level of visitor interaction with the service process. When both of these levels are high, visitors actively participate in the service process and the business works to satisfy the visitors' particular preferences.

For enterprises with low ratios of labour to capital intensity—that is, Service Factory and Service Shop processes—physical capacity decisions are paramount. Here, capacity cannot be augmented easily, so it is essential to utilize each capacity unit effectively (each airline seat, hotel room, restaurant or gaming table seat, amusement park ride seat, etc.). Strategic physical capacity decisions encompass the long-term planning of construction or renovation of buildings, planes, or cruise ships, or the purchasing of new equipment, rides, vehicles, etc. Short-term capacity management, in contrast, includes approaches that enable firms to react to short-term fluctuations in demand. The latter is typically achieved through scheduling, yield or revenue management, and the addition or withdrawal of support services.

Mass Service and Professional Service processes, on the other hand, have high degrees of labour intensity. Strategic capacity management for these enterprises focuses more on the management of labour capacity, including the hiring and selection of employees with the appropriate skill mix. Short-term capacity management focuses on the implementation of scheduling and

other incremental capacity methods that allow the enterprise to react to short-term fluctuations in demand with greater flexibility.

3. Capacity measurement

Capacity measures serve as the foundation of all analytical approaches for all capacity decisions. For example, a capacity measure could provide the planning objective (e.g., an 80% employee utilization goal). Or the measures could show the relationship of capacity to the overall performance of a system (e.g., at 80% and 90% maximum capacity, a park experienced 89% and 75% customer satisfaction ratings, respectively). But as pointed out by Wall (1982), capacities are difficult to measure because of definitional problems. The appropriate design and outcome measures depend on whose perspective is emphasized. In Table 2, we provide some examples of operational indicators that are particularly relevant to managers and visitors.

In most for-profit environments, management is concerned with maximizing revenue through the number of paying visitors using the enterprise or revenues generated per visitor and the firm's capacity utilization of resources. Visitor objectives (including less crowding and shorter perceived waiting times) and their other specific measures of relevance differ from those of the enterprise management. In hospitality and tourism, both biophysical (the quality of the environment) and behavioural components are recognized (Wall, 1982). An increase in physical capacity utilization can lead to queuing and a general reduction in the perceived quality of service (Davis and Heineke, 2005). In the mainstream literature, the critical levels that are needed to support certain facilities are often referred to as capacity thresholds (Saveriades, 2000). To cite an example, high occupancy of an airplane or a quick-service restaurant can lead to untidiness, as the service crew simply does not have time to maintain cleanliness; this is a negative from the visitors' perspective (Muller, 1999). A nearly empty cruise liner, on the other hand, does not provide the expected social ambience and affects the employees' interactions with customers (Kandampully, 2000).

Clearly, managers must monitor visitors' perceptions and preferences and try to best meet both management's and their customers' expectations. For example, an Australian casino improved the level of patrons' comfort by decreasing the playing spots offered on each gaming table, which subsequently increased the blackjack hold (Kilby and Fox, 1998). However, as Muller (1999) illustrates, a reduction in the number of customers can be compensated for by higher revenues only if the demand is price inelastic.

Different capacity indicators linking the ability of an enterprise to control the environment with the overall customer experience can be used. For example, in a restaurant setting, capacity-related indicators include average customer waiting time and average duration of each service. These indicators have been shown to have

Table 1
Application of the service process matrix^a to the field of hospitality and tourism.

Degree of labour intensity	Degree of interaction and customization	
	Low	High
Low	<i>Service Factory</i>	<i>Service Shop</i>
	Airlines	Luxury hotel
	Standard hotels	Luxury restaurants
	Resorts	Luxury cruise ships
	Cruise ships	Spas
	Recreation	
High	Theme parks	
	<i>Mass Service</i>	<i>Professional Services</i>
	Retailing	Tour guides
	Cafeterias	Instructors
	Fast food	Concierges

^a Schmenner (1986).

Table 2
Examples of capacity measures relevant to visitors' versus managers' perspectives.

Visitors' perspective	Managers' perspective
Availability	Revenue or profits
Right time	Number of visitors
Right price	Utilization
Right type	Percentage or cost of depletion or wear
Crowding and space	Excess capacity or waste
Too many people/too little space	Labour costs
Too few people/too much space	Variable and fixed physical resources
Intimacy/privacy	Actual service time
Interaction/sociability	Actual wait time
Perceived service time	
Perceived wait time	

a causal relationship with the number of incidents/complaints, the general ratings of the service, and the food quality (Adenso-Díaz et al., 2002). For a recreational site, capacity indicators can include the following: the satisfaction of the visiting public, the intention to return, the percentage of people feeling overcrowded, and the number of complaints (Simón et al., 2004). Several researchers have shown that the links between capacity decisions and other performance measures such as profit, quality (overall visitor satisfaction), and market share are complex and nonlinear (Easton and Pullman, 2001; Sill, 2000).

4. Strategic physical capacity decisions

As mentioned previously, strategic capacity planning of physical resources is of primary concern for Service Factory and Service Shop processes (hotels, resorts, cruise ships, recreational sites and theme parks). In practice, this planning is part of the detailed design that includes capacity estimation, an integral part of a feasibility study (Lawson, 1995, p. 3) and the main indicator of the price of a capital asset (Roubi and Litteljohn, 2004). Strategic physical capacity decisions include size, configuration, layout, and built-in flexibility. These decisions made during the planning stage have long-term implications for the overall resource productivity of a property (Brown and Dev, 1999).

The capacity-related criteria for the “goodness” of a design include the grossing factor (Ransley and Ingram, 2001), which is calculated as the proportion of income generating areas relative to the total floor space. The grossing factor is reduced by the need to provide technical areas (business centres, sport and recreational areas, etc.) and functional features (architectural elements such as atriums, balconies, and grand staircases), as well as by the need to meet legislative and environmental requirements such as land zoning and usage restrictions, building codes, fire and maritime safety requirements, and disability access regulations.

Codes for food-service premises, for example, stipulate the allocation of adequate space for each food production area and regulate the physical layout of the facility to ensure the ease of cleaning/sanitation (minimum gaps between equipment units and

walls, minimum number of hand-washing sinks and maximum walking distance to reach them, etc.). These requirements for land-based (Standards Australia, 1998; U.S. Food and Drug Administration, 2005) and cruise-ship (Center for Disease Control and Prevention, 2001) kitchens lead to an increase in the size of the production areas, which are non-revenue-generating. By means of the grossing factor, various areas of a facility can be assessed for the contribution margin of the different service products they support. For example, in the Raffles City Convention Centre in Singapore, the contribution margin is 30–35% for food services, 85–95% for room rentals, and 50–95% for audio-visual equipment (Kimes and McGuire, 2001).

In addition to size, the actual physical capacity depends on the configuration and layout of a facility, since the combination of these affects the flow of products and customers. For example, in restaurant planning, table types and locations (banquettes, booths, or anchored or unanchored tables) affect the customer experience and, hence, the average spending per minute (Kimes and Robson, 2004; Robson, 1999). Examples of other design solutions for different sectors are given in Table 3. Architects and designers use computer-based tools such as computer-aided design (CAD) to “visualize” projects during the conceptualization stage. The service context is reflected in the fact that other considerations often take precedence over the principles of space efficiency and rational product/customer flow. In the cruising sector, for example, social stratification rules and the stability of different sections of a ship against rolling determine the positioning of different categories of cabins (Junko, 2004).

Built-in flexibility addresses another key strategic physical capacity issue, since it enables future expansion or changes to adapt to market demand. The foundations of a building may be constructed for horizontal or vertical expansion in the future, its walls collapsible or removable to accommodate different type of activities. Flexibility and modularity principles are also applicable to furniture and equipment design. In the cruising sector, on the Queen Mary 2 (Cunard Cruise Liner) for example, the restaurant décor changes during the day, as a self-service breakfast café becomes a lunch-time café-bar and then a waiter-served evening bistro. On Carnival Cruise Lines, modular beds and furnishings are

Table 3
Examples of layout approaches for physical resources.

Industry	Layout solution	Outcomes	Reference
Airlines, theme parks, ski resorts	Hub and spoke system	The hub is a central facility offering restaurants, shopping arcades, entertainment, conference rooms, and other amenities, while the spokes are chosen destination areas.	Wanhill (2003, p. 47)
	Point to point	The space is designed for people to move from one destination area to the next without moving through a central location.	Hanlon (1989)
	Virtual queue	Customers sign up for virtual queue, go to other destination areas and return at appointed time.	Dickson et al. (2005)
Hotels	Increased bathroom size	The decrease in the net bedroom size coupled with an addition of a walk-in shower and dressing area resulted in a 13% improvement of the design efficiency ratio.	Ransley and Ingram (2001)
Restaurants	Process flow principle	A steady, controlled process flow reduced complexity and confusion in a kitchen. Better distribution of serving capacity between cold and hot sections in restaurant buffets resulted in faster table turns, shorter lines, better food temperature control, and decreased construction costs.	Brown and Hoover (1990) Sill (2000)
	Table type and location	Customers in booths spent more on average than those in banquettes; those in poor table locations spent more.	Kimes and Robson (2004)
	Reduce table spacing	Creating a busy feel (small dining spaces) increased customer satisfaction.	
Cruise ships	Horizontal class sandwich	Positioning of the upper category of cabins at the lower desk to minimize rolling, which is amplified at the upper deck as well as at the bow and the aft.	Junko (2004)
	“Cocooned isolation”	Vertical class barrier with the provision of a view, a private balcony, and no need to change floors to reach the nearest pool.	Junko (2004)

used in standard cabins so that rooms can be easily configured as a set of twin beds or a queen bed depending on demand type.

Food production is the only manufacturing function in the hospitality sector (Jones and Lockwood, 2002), but it faces more unpredictable demand patterns than most goods production. However, this can be managed with the use of technology by choosing suitable food service systems. Short shelf-life systems (cook-hot-hold and cook-chill) offer a relatively small time buffer between food preparation and consumption (Creed, 2001) and are suitable for settings with predictable demand, such as banqueting or in-flight catering. Extended shelf-life systems (cooked-in-a-bag and cook-freeze), on the other hand, which allow for a longer time buffer, are suitable for unpredictable demand patterns such as those of railway food services or restaurants that are open to the public. The practical implication of this for tourism-oriented food services is that the capacity of their physical resources can be better utilized through the production of packaged meals for new markets (local cafés and restaurants as well as the retail sector) during a slow season or periods of tourism downturn.

5. Strategic physical capacity estimation

The relationships between demand, capacity, managerial objectives, and the visitor experience are complex and difficult to evaluate without appropriate analytical techniques. In Table 4 we list different methods used in strategic capacity estimation. The range of strategic estimation approaches range from the planning of overall tourism or hotel carrying capacity in a city (for long-range tourism or special events such as the Olympics) to such micro-level decisions as the sizing of hotel water heaters.

5.1. Closed-form approaches for physical capacity estimation

Generally, the major capacity indicators (the number of berths on a cruise ship, rooms in a hotel, seats in a restaurant, etc.) are determined during the initial stages of major infrastructure projects. These indicators are then used later in the estimation of other capacity elements. Closed-form/cost-oriented methods are typically used in the accommodation sector, which is capital-intensive, since the capital costs, such as construction costs, furnishings, property taxes, and insurance, vary with capacity. An example of a closed-form solution is a typical break-even formula such as the cost-volume-profit method (Liu and Var, 1980) where:

$$P = Q(S - V) - F$$

where P = total profit; Q = volume of sales (e.g., visitor-days); S = sale price per unit volume; V = variable cost per unit volume; F = total fixed cost.

Here, if a target profit is set, the volume of sales (Q) can be derived from this equation, which together with the expected occupancy level would indicate the capacity needed to meet the given profit objective. With this approach, the capacity depends on the targeted sales price. Another closed-form approach, proposed by Carey (1989) uses regression models to determine overall hotel carrying capacity for specific locations. Here the dependent variable, optimal room occupancy is a function of the independent variables, price and overall market capacity.

A feasibility assessment of a new project also includes the determination of variable or semi-variable costs affected by furniture/equipment needs and layouts. These can be classified as “capacity elements” and are usually estimated using empirical formulae with results expressed in a variety of units (square meters for different functional areas; number of parking places/rides/boats, etc.; customers per lift cycle for a ski resort; standing spots for casino games; and others). These approaches are usually simpler than the methods used in determining the major capacity indicators and reflect the factors shaping the capacity on the micro-level, which are usually not directly linked to profits. For example, Muller (1999) provides a very simple empirical formula to estimate capacity for a restaurant based on estimates of hours of service, number of seats, cover count, and planned service-cycle time:

$$\text{maximum seating capacity} = \frac{\text{seats} \times \text{hours of service}}{\text{service} - \text{cycle time}}$$

The closer a capacity element is to the customer interface in a physical sense, the more customer-related variables are used in capacity estimation—the number of customers served per hour for a dining-room or the percentage of people travelling by car for parking, for example. To estimate the number of rides in a theme park, Wanhill (2003, p. 45) provides a closed-form solution integrating the catchment’s population and penetration rate, demand fluctuations between weekdays and weekend, average “consumption” of entertainment units per customer, and average ride throughput.

5.2. Probabilistic methods

Typically closed-form solutions must assume that variables such as pricing or demand are constant or that certain conditions must be ignored to create problem tractability. As an alternative, probabilistic models allow for more complexity and realism. These approaches range from modifications of closed-form approaches to large simulation models. For an example of a modified closed-form solution, Gu (2003) treated the supply of room nights of Las Vegas

Table 4
Examples of strategic physical capacity estimation techniques.

Sector	Units	Method	Reference
Hotels	Beds, rooms, percentage of occupancy	Cost-volume-profit Regression analysis Economic order quantity	Liu and Var (1980) Carey (1989) Gu (2003)
Restaurants	Available seats per hour Degree of table “combinability” Production equipment units	Number of seats $\times$ hours of service/service-cycle time TABLEMIX computer program Empirical formulae	Muller (1999) Thompson (2002) Berdichevskiy and Kapsekin (1988)
Gaming	Average blackjack hands/hour	Minimum break-even bets	Kilby and Fox (1998, pp. 198–226)
Theme parks and ski resorts	Rides Ski lifts	Break-even analysis Simulation modelling	Wanhill (2003, p. 50) Pullman and Thompson (2003)
Convention and exhibition centres	Square meters of exhibition area, tables, flatware, chairs	Benchmarking against the consumption patterns (number of customers per event, space, time requirements, etc.) and future trends in demand	Schnitzler (1982)
Cruise ships	Berths	Dynamic game theory	Wie (2005)

Strip casino hotels as inventory units and integrated a probabilistic model based on the economic order quantity (EOQ) equation to reflect the perishable nature of hotel services. This approach allowed for the prediction of the cycles of over- and under-capacity. Another highly sophisticated approach, the open-loop Nash equilibrium, has been used to quantify the effects of the number of berths on the profits of a hypothetical cruise liner under oligopolistic conditions (Wie, 2005).

Discrete-event simulation models typically rely on software packages such as *Process Model* (2005) to model dynamic and complex conditions. Any capacity problem can be modelled using simulation software with the only constraint on realism being the availability of actual data and modelling skills of the developer. In previous research, simulation modelling has been applied to capacity problems such as restaurants, cruise ships and ski resorts. For example, Wall (1982) uses simulation modelling to account for multiple inputs and the cyclic nature of capacity in tourism. For restaurant capacity, the TABLEMIX model developed by Thompson (2002) provides the optimal percentage of combinable tables in a restaurant by integrating the number and duration of peak dining periods, the probabilities of different numbers of party arrivals for the dining period, the probabilities of different size parties, the maximum number of waiting parties, the distribution of dining periods (normal or lognormal), and the choice of table arrangement rule (e.g., assign an available table to the largest party or to the party waiting the longest).

Using probabilistic models of market share and computer simulation to more accurately integrate the relationship between capacity and demand, Pullman and Thompson (2003) developed a method for determining the profit-maximizing capacity strategy for different hospitality environments. This method requires customer utility models (reflecting real customer's preferences for multiple capacity-related aspects of a service such as queue duration) with the potential costs and revenues from the implementation of different capacity management strategies. Using discrete-event simulation models of a resort's subsequent flow performance under different capacity scenarios, the profit-maximizing strategy simultaneously addresses both management and customer preferences as shown below:

$$\text{profit} = \sum_{t \in T} (N_t \pi_{jt} (P_t - V_j) - F_j)$$

where t = time period (set T periods); N_t = number of potential visitors in period t ; P_{jt} = price of service j during time period t ; V = variable cost per visitor for service j ; F = fixed cost of service j ; π_{jt} = potential market share or probability that a customer group will select service j from the set of J services in period t .

The above market share is derived from actual customer preference models based on choice experiments (please see Verma et al., 2002 for more details). Here, the probability that an aggregate group chooses service j out of a set of J competitive service establishments is the following:

$$\pi_j = \frac{e^{U_j}}{\sum_{j=1}^J e^{U_j}}$$

where U_j = aggregate customers' overall systematic utility for service design j .

The advantages of these mixed-model approaches are as follows: first, complex enterprises can be modelled with a great deal of realism, including variable costs, dynamic customer arrivals and flow patterns, multiple combinations of strategies, and heterogeneous customer mixes. Alternate forecasts of visitor behaviour and customer flows can be tested. Second, utility models are typically based on actual customer preferences for different attributes of an enterprise (for example, crowding level, average queue time, price, retail and amenities level, mix of

activities, or skill level). Third, by integrating the customer and enterprise perspectives, diverse criteria can be optimized, including profit, revenue, and customer satisfaction. This approach allows for the testing of various capacity management strategies, enabling management to evaluate their estimated costs prior to the capital investment.

6. Strategic labour capacity decisions and methodologies

For enterprises, strategic labour capacity decisions cover a long-term planning horizon, 6–18 months (Schroeder, 2006). As with physical capacity decisions, firms' functional capacity strategies should reflect their competitive priorities. From an operations management perspective, there are three typical strategies for labour capacity planning: level, chase, or mixed (Sasser, 1976). For example, hotels with a high service/competitive priority might want to retain the majority of their staff throughout the year (a level strategy), regardless of demand, to keep both the service level and the morale high. A cost-focused hotel, on the other hand, might only staff as needed (a chase strategy) to minimize labour costs.

A level strategy is more appropriate when demand is known before the time of use, the firm has strict requirements about service availability, and/or the employees are scheduled according to pre-existing criteria such as union contracts (Sasser, 1976). Typically, Service Factory processes with low interaction, customization, and labour intensity use this strategy. Demand is scheduled to fit into an existing plan, such as airline flight or tourist bus schedules. The visitor or guest reserves a space in advance and the labour capacity is fixed for the flight or bus trip regardless of the number of people on that trip. This approach is best suited to an environment in which labour requirements are predetermined based on physical capacity, such as under safety regulations (e.g., the number of flight attendants on a plane is fixed regardless of the number of passengers).

A chase strategy tries to meet demand as it occurs (Sasser, 1976). For this type of strategy, the labour requirements change to match the demand forecast throughout the planning horizon. This approach is best suited to an environment in which visitors cannot be prescheduled and there is a high degree of labour interaction and customization (Professional Services), or to other environments with unpredictable visitor arrivals such as hotel front desks, shops, parks, and non-reservation restaurants.

Mixed strategies use some combination of level and chase approaches (Sasser, 1976). Because of the seasonality of many tourism enterprises, it makes sense to use level strategies for certain staff areas (e.g., cruise ship operators and maintenance or other departments that must function regardless of the number of visitors) and chase strategies for others (maid service, ski instructors, front desk personnel, wait-staff, bell-staff, etc.).

Often a leisure enterprise will change its labour capacity strategy as the company grows. For example, Port Aventura, a Spanish theme park, used a chase strategy for their first season to match employees to demand. During that season, the workforce reached 2800 on the busiest days but went down to 200 employees during the winter season. Later, the company signed a collective agreement with their unions that offered permanent or permanent-seasonal contracts to 50% of the long-term workforce, thus requiring a mixed strategy (Huete and Segarra, 1997).

Labour capacity estimation for various strategies depends on the industry and often on the planned physical capacity. Here, industry standards play a significant role, dictating how many employees of each type are required for a plane, restaurant or hotel of a certain size. Sill (1991) provides examples of this approach for hospitality. An enterprise first estimates a key demand input (e.g., number of expected guests, menu-mix history, or potential bar sales), multiplies that by a given standard (staffing standard,

recipe, table-setting used) and gets a capacity output (staffing for the dining-room, kitchen, dishwashing, food production, or bar). To cite another example, in the hotel industry, union contracts might dictate how many rooms a housekeeper may clean in a shift (perhaps 12–14 standard rooms), and the required number of housekeepers can then be directly calculated from the expected room occupancy.

For capacity estimation that involves a partial or full chase strategy, a typical approach to labour staffing for long-term planning employs aggregate planning techniques that calculate the economic trade-off between full- and part-time employees, overtime, and hiring and lay-off costs to determine the labour requirements. These problems are solved optimally using linear programming in which the objective is to minimize cost or maximize profit subject to constraints such as the meeting of demand, limits on materials and equipment time, and labour requirements for standard and overtime wages (Bozarth and Handfield, 2006, p. 396). In a typical inventory-less model formulation, the linear objective function for cost minimization is the following (Vollman et al., 2005, p. 413):

$$\min \sum_{t=1}^M (C_H H_t + C_F F_t + C_R X_t + C_O O_t + C_U U_t)$$

subject to the constraints that

- regular-time hours must not exceed the allowable maximum to be worked per employee per month;
- workforce level change between months $\geq H_t - F_t$; and
- $X_t - O_t \geq D_t$

where C_H , C_F , C_R , C_O , and C_U = the cost of hiring an employee, the cost of firing an employee, the cost per labour-hour of regular time, the cost per labour-hour of overtime, and the cost per labour-hour of idle regular time, respectively; H_t , F_t , X_t , O_t , and U_t = the number of employees hired, employees fired, regular-time hours, overtime hours, and idle regular-time hours in month t , respectively; D_t = the number of hours of service to be sold in month t ; and M = the number of months in the planning horizon.

Similar linear programming models can be set up to model more complex environments, such as different product or customer classes, different employee classes (part-time and full-time), and many other service situations (Vollman et al., 2005, pp. 415–423).

7. Short-term capacity decisions and methodologies

Short-term capacity decisions involve the day-to-day attempts to match capacity to demand. Many hospitality and service researchers have addressed these types of decisions and estimation techniques, since these management techniques have the greatest potential to immediately improve the visitor experience (Adenso-Díaz et al., 2002; Klassen and Rohleder, 2002; Sill, 1991). In this section, we will again differentiate between the physical and human (including labour and visitor) capacity approaches.

7.1. Physical capacity

With the exception of certain Professional Services, all types of sectors can benefit from physical capacity approaches. During the actual operation of an enterprise, the ability to meet demand depends on both the flexibility of the physical resources and the willingness of the price/market segment to pay for an available capacity unit.

In Table 5 we provide a comprehensive list of the different physical resource approaches (Adenso-Díaz et al., 2002; Klassen

Table 5

Short-term capacity management approaches.

Physical capacity
Physical flexibility
Rent capacity
Share capacity
Hire sub-contractors
Change resource allocations
Change hours of operation
Provide off-site access
Use automation
Price/segment flexibility
Partition visitors (status and length of transaction)
Yield management
Revenue management
Human capacity
Labour flexibility
Schedule employees
Allow overtime
Allow idle time
Cross-train employees
Change work speed and process
Hire permanent employees
Lay-off employees
Use temporary employees
Use part-time employees
Visitor flexibility
Allow waiting
Allow balking
Turn away visitors
Provide rewards or incentives
Provide diversions or complementary services
Camouflage the queue
Pay for VIP queues
Change level of visitor participation
Schedule visitors/take reservations
Inform/educate about alternative options

and Rohleder, 2002). Most of these physical flexibility approaches are self-explanatory. The majority of price/segment flexibility approaches use capacity partitioning or yield/revenue management techniques. Yield and revenue management methods are typically defined as selling the right product to the right customer at the right price and for the right duration (Kimes, 1989; Kimes et al., 1999; Weatherford and Bodily, 1992). Yield management does not result in any additional goods or service capacity; rather, it links markets with an existing company's physical and operational resources. Traditionally, yield management has been used by airlines (price and class), lodging enterprises (price and stay duration), restaurants (price, style, and duration of service/meal), and more recently, by gaming venues (gaming spots per table and minimum bets) and the golf-course industry (no-show fees, rental fees, playing times). The most basic methodology is the linear modelling approach. Here the objective is to maximize revenues from different customer classes subject to the capacity constraints of the enterprise as illustrated below:

$$\max \sum_{i=1}^n r_i x_i \text{ subject to } \sum_{i=1}^n x_i \leq c \text{ and } x_i \leq d_i$$

where i = rate class or segment, x_i = number of items sold in rate class i , d_i = demand for rate class i , r_i = revenue from selling items in rate class i , and c = capacity of the service.

Yield management methodologies and research have been expanded to include probabilistic demand variables, economic approaches, network approaches, customer satisfaction implications, etc. (e.g., Lindenmeier and Tscheulin, 2008; Sanchez and Satir, 2005). In addition, yield management has been applied to almost all leisure and hospitality sectors. For examples of the various short-term capacity estimation techniques used in different sectors, see Table 6.

Table 6

Capacity estimation in different hospitality and tourism sectors.

Sector	Tactics	References
Hotel/resort	Packages: budget, moderate, and deluxe Categories: couples, singles, and families Subcategories for children: kids, petit, and mini baby/nursery	Kandampully (2000)
Restaurant	Off-peak discounts Increase production rate by application of more efficient equipment and simpler preparation techniques or computer-based calculations and ordering Reduce randomness via well-designed reservation system Increase sophistication through application of queuing theories, modelling and simulations, linear programming, and forecasting In low-demand periods: assign multiple duties to the host, increase menu variety, offer regular or reduced prices, increase promotion, have servers bring condiments	Muller (1999) Kimes et al. (1999)
Ski resort	Capacity upgrades: improved lifts Capacity expansion: additional terrain Queuing information Inter-daily demand smoothing: surcharge during weekends Class variation: mix of beginners and intermediate	Pullman and Thompson (2003)
Golf course	Reduce the uncertainty of arrival by means of forecasting, overbooking, requiring guaranteed reservations, reconfirming reservations by phone, instituting a no-show fee and/or a fixed fee for reservations Reduce the uncertainty of duration by selling rounds of predictable length, flexible course design with a global positioning system to direct flow, requiring own golf carts or reducing the cart rental fee during busy times, using course marshals and caddies, posting golfers' playing times Reduce the tee-time interval	Kimes (2000)
Small Inn/B&B	Coping: close part of year, maintain the premises, 'reduce staff' seek other income, borrow money Combating: stay open all year, develop other tourist segments, cater to residents, augment the product's appeal, add value, develop export products Capitulating: shrink, sell, or terminate the business	Getz and Nilsson (2004)

The capacity requirements of different customer groups cannot only affect the revenue but the overall utilization of a facility. For example, to increase utilization, a ski resort with a large percentage of terrain for expert or advanced skiers increased its promotions for the "family skier" segment (skiers with beginner-level and intermediate skills). This change in customer mix was shown to degrade the service quality in terms of queuing but resulted in increased spending on lessons and amenities (Pullman and Thompson, 2003). In a golf course setting, experienced golfers usually play faster than novices. Offering separate playing times for beginners relieved the experienced players' frustration and improved overall flow (Kimes, 2000). For outdoor recreation enterprises, the attraction of environmentally aware tourists in order to minimize ecological damage and increase tourism carrying capacity (Simón et al., 2004) had a similar effect on the operations.

7.2. Human capacity

Tactical human capacity approaches involve labour and/or customer flexibility. From the labour perspective (Table 5), most of the flexibility comes from hiring an appropriate number of employees, scheduling employees to meet demand as it occurs, and adequately training employees to do their respective jobs or cross-training them to do multiple jobs. The majority of these labour-related tactical capacity management approaches derive from the operations area and are based on queuing theory. According to Thompson (1998a), matching visitor demand to labour capacity involves a four-step process: (1) forecasting demand; (2) translating the demand forecasts into requirements for employees; (3) developing a workforce schedule that, ideally, only has employees working when necessary to deliver the service; and (4) controlling the delivery of the service in real time. While demand forecasting itself is beyond the scope of this paper, we

note that many researchers have examined the remaining three steps, which relate to the use of these forecasts by the enterprise.

From the customer perspective, flexibility tactics involve the management of waiting lines or "capacity buffers." The process can be designed to allow visitors to merely wait; to let them leave rather than keeping them "captive" in roped areas; to provide diversions to improve the line experience; to hide the queue; to offer higher prices in exchange for a shorter queue; or to allow virtual queuing (e.g., electronic queuing, such as Disney's Fast Pass system). Using queuing theory and the appropriate model, analysts can determine the appropriate number of workers based on line configuration, desired length of lines, customer wait duration, and average service times (see, e.g., Kleinrock, 1975). Alternatively, visitors can increase their own level of participation in the process, allowing the enterprise to increase the available labour capacity. For example, a self-guided audio-tour system allows museums, parks, and other tourism sites to reduce the number of guides required.

A wide range of labour capacity techniques are available that address these problems. Thompson (1998a,b, 1999a,b) offered a comprehensive tutorial on the development of optimal workforce schedules for multiple employee classifications (part-time, breaks, and reliefs). Generally, the goal of Thompson's approach is to minimize employee cost while meeting customer service-time goals. Adenso-Díaz et al. (2002) offered an alternative approach that establishes the minimum labour staffing levels below which quality may be affected. In their study, quality was measured according to visitors' perceptions (see Table 2), based on historical data in a catering sector. Along those lines, Goodale et al. (2003) illustrated a method that schedules labour to maximize profit (creating the most desirable quick-service restaurant configurations from both the visitors' and management's perspectives) using consumer utility-based models in an international airport food-court. Both of the latter labour-scheduling methods have the

Table 7
Capacity estimation approaches and their application.

Type	Description	Time frame	Purpose	Limitations	Advantages
Physical capacity: statistical methods	Method uses historic data with regression and/or probabilistic models to predict requirements.	Performed during the planning phase of the project.	To estimate hotel carrying capacity or room occupancy.	Requires historic data for the specific location. Difficult to extrapolate to new locations.	Allows for cycles of over- and under-capacity.
Physical capacity: closed-form/cost focus	Method weighs cost, volume and profit to determine break-even capacity.	Performed during the planning phase of project.	To estimate the number of capacity units (berths on ships, seats in plane or restaurant, hotel rooms, etc.) based on desired profit and costs.	Requires predictions of prices and costs. Does not account for competitive environment and economic fluctuations.	Relatively simple approach with closed-form solution.
Physical capacity: mixed statistical and customer preference models	Method weighs customer preferences, competitive market, and simulation of enterprise scenarios.	Performed during the planning phase of project.	To estimate appropriate capacity based on customer preferences for different attributes of the proposed enterprise in a competitive market.	Requires market research on customer preferences and existing or proposed competitors along with modelling of enterprise function.	Allows for most realistic view of multiple performance measures under any number of future scenarios.
Physical capacity: simulation models	Method models the proposed facility under various conditions.	Performed during the planning phase of the project.	To simulate the customer flow and queuing under different levels of capacity flexibility, layout, and capacity choices.	Requires sophisticated modelling skills, projected demand data, software, and complete understanding of operating environment and rules.	Allows for high degree of realism and multiple scenario evaluations. Can test design changes of existing enterprises without disrupting flow.
Physical capacity: yield management	Estimate uses historical and current demand to determine optimal capacity use.	Performed as needed, depending on the demand changes for the enterprise.	To estimate the right unit of capacity to sell to the right customer at the right price and for the right duration.	Requires historical data, programming ability or specialized software.	Determines optimal price, customer class size, and capacity.
Labour capacity: deterministic	Method links labour requirement to physical capacity units or regulations.	Performed after physical capacity is determined.	To estimate the number of employees based on heuristic or industry regulations.	May not allow for flexibility or adaptation to actual needs.	Relatively simple to implement.
Labour capacity: linear programming	Method finds minimal labour requirements based on fluctuating demand and supply conditions.	Performed as needed (daily, weekly, monthly) depending on the industry.	To estimate the number of employees needed, given projected demand, employee service rate, contract schedule requirements, and constraints on service level.	Requires programming ability or specialized software. Can be difficult to account for different employee skills or productivity levels.	Determines optimal cost and profit schedules.
Labour or physical capacity: customer flexibility tactics	Methods allow customer to chose service method.	Performed during enterprise operations.	To free up capacity by allowing customer to self-serve or choose virtual lines.	Requires an additional channel for self-service or a virtual queue system.	Allows customers to choose their experience and reduces capacity costs.

ability to include multiple measures of visitor-preferences (such as crowding level, visual displays, layouts, and food variety) in addition to capacity costs and revenues.

Klassen and Rohleder (2002) ran multiple simulated scenarios that combined a number of labour and visitor flexibility options simultaneously to determine the best combinations of approaches. Their results indicate that enterprises with limited flexibility in terms of maximum schedulable staff should make use of customer participation, employee cross-training, and price segmentation and should inform or educate customers about other alternatives.

Finally, the real-time adjustment of labour schedules (Thompson, 1999b) refers to methods in which managers observe actual visitor demand and correct any capacity problems as they occur. Here the goal is to improve service and cost performance in real time. For example, if a restaurant's customer flow is slower than expected, employees can go on break or leave early; if it is busier than expected, employees can defer breaks until a later point in their schedule. Real-time schedule adjustments include changing employees' station assignments, cancelling or changing their breaks, starting their shifts early/late or asking them to leave early/late, calling in more workers, or cancelling shifts (Hur et al., 2004). Thompson (1999b) indicated that schedules developed with preset employee breaks outperformed those with real-time break adjustment in terms of overall cost. However, in their work, Hur et al. (2004) found that the active adjustment of work schedules is beneficial as long as the direction of demand change is accurately identified.

8. Conclusions and implications

In this paper, we have sought to differentiate between long-term strategies and short-term tactics in capacity management, and we have provided a review of current research and methodologies in this area. Several interesting insights are brought to light by this review. First, most of the recent capacity management research has been case-focused (e.g., Gu, 2003; Saveriades, 2000; Simón et al., 2004) or industry-specific (e.g., Pullman and Thompson, 2003; Wie, 2005). There seems to be a common thought that each situation is unique because its capacity management is sensitive to the context, the speed of visitor/guest growth, and the dynamic nature of supporting resources (technology, etc.). However, we have found that there is much to be gained from sharing capacity management strategies and tactics across industries, particularly given certain common characteristics: the influence of nonoperational factors on capacity, the dynamic nature of problems, and the existence of multiple stakeholder perspectives and measures.

It can be concluded that capacity planning approaches are based on a wide spectrum of techniques. The closed-form techniques are generally not sensitive to the impact of changing external factors and the dynamic nature of supply and demand. The break-even analysis, for example, does not take into account the change of price with volume or the change of product mix or economic conditions over time. Another problem is the difficulty of disaggregating fixed and variable costs in practice. Regression analysis and consumer preference models, on the other hand, allow for the incorporation of complex internal and external effects, many of which are intangible. Computer modelling offers one possibility for simulating the variability and interrelationships of the systems. Current software packages (e.g., *Process Model* (2005)) have graphic-output capabilities to describe typical settings such as restaurants, airports, cruise ships, and theme parks. A detailed analysis of the advantages and disadvantages of various physical and labour capacity estimation approaches is provided in Table 7. Main criteria include the simplicity of use, the modelling skills and amount of data needed, and the adaptability of

the approach to different environments and over- and under-capacity cycles.

Capacity design solutions are affected by a number of nonoperational factors such as social stratification rules. Simulation models that address these issues are now being developed and applied to these problems, and the industry is facing a significant opportunity to incorporate more sophisticated capacity management tactics across different enterprises. Traditional yield management approaches, for instance, which are based on such concepts as customer demand and the ability to segment by rate class, have been widely adopted by the hospitality industry; however, thus far these approaches have received limited attention from leisure enterprises such as parks and other recreational sites. Yield management style models could be adjusted to supplement price with non-price attributes (crowding, wildlife, safety, etc.) so that visitor "utility maximization" occurs subject to the capacity constraints of the environment.

Conversely, the carrying capacity models currently used for tourism sites could yield benefits for other hospitality environments. For example, the recent trend in boutique hotel design alters the traditional lobby model to merge the cocktail lounge/disco and guest check-in/entry location (Appleyard, 2002). The properties fill the lobby to maximum physical capacity without considering the carrying capacity. But when hotel guests struggle through crowds to check-in or to make use of the facilities, the quality of the hotel customers' experience is sacrificed for the sake of food and beverage revenue maximization.

A limited number of studies have explored multiple stakeholder capacity optimization models using customer preferences (Pullman and Thompson, 2003; Goodale et al., 2003; Adenso-Díaz et al., 2002). The tactic of combining several techniques (e.g., cost-based models, regression analysis or consumer utility models, with simulation) provides a holistic approach that integrates numerous inputs and the variability of service products. While the yield/revenue management approach has been applied to multiple sectors of hospitality, other mixed models have not received the same level of broad application. So far, these models have been applied to ski resorts, airport food-courts, and restaurant environments, but they also hold much potential for other types of enterprises. Practitioners in particular can benefit from models that integrate consumer behaviour with capacity attributes.

As we have seen, service capacity utilization has a significant influence on visitors' perceptions of quality: both over- and under-utilization can be undesirable. Models that include both quality and productivity measures may enable enterprises to determine optimal physical as well as social carrying capacity for a variety of environments.

This review has several limitations yet offers opportunities for further research. First of all, we cover existing research on a wide range of capacity methodologies. The breadth of the topic, however, limits the extent to which we are able to address individual capacity management methods in great detail. Second, capacity management requires a multidisciplinary vision and solutions. We have attempted to review a wide cross-section of research in the areas of tourism, hospitality, marketing, and operations, but it is impossible to include all the papers on this subject. Further research into the application of alternative capacity methods to untested areas is essential to the improvement of both the strategic planning and implementation processes.

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